MILITARY SPECIFICATION

CARTRIDGE, PHOTOFLASH, M123A1, PARTS FOR, AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the parts and the loading, assembling and packing for one type of photoflash cartridge designated as Cartridge, Photoflash, M123A1.

1.2 Classification. The photoflash cartridges shall be of the following classes:

Class 1 - 2 Second Delay
Class 2 - 4 Second Delay
Class 3 - 6 Second Delay

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-P-223	- Powder, Black
MIL-P-11329	- Primer, Electric, M59, LA & P
MIL-A-48078	- Ammunitions, Standard Quality Assurance Provisions, General Specification For

FSC 1370

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn. DRSMC-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-151 - Metal; Test Method

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1234 - Pyrotechnics, Sampling, Inspection and Testing

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER (ARDC)

7548472 - Carton, Packing, Ammunition, for Cartridges, Photoflash, M123A1 and M124
7548473 - Box, Packing, Ammunition, for Cartridges, Photoflash, M123A1 and M124
8847519 - Cartridge, Photoflash, M123A1, 2 Sec., 4. Sec., and 6 Sec., Delay, Assembly
19-48-4116/106A - Unitization Procedure for Box Ammunition and Components on Four Way Entry Pallets Cartridge, Photoflash M123A1

INSPECTION EQUIPMENT DRAWINGS

9278637 - Procedure for Light Output Measurements of Photoflash Cartridges

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (dwg.) 8847519 and with all requirements specified in applicable specifications.

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3.3 Static Illumination of Charge Case Assemblies.

3.3.1 Dud. The charge case assembly shall not fail to function.

3.3.2 Integral light value. The total integral light value (sum of end on plus side on light readings) of each charge case loading assembly during the first 0.040 seconds of flash shall be not less than 8 million candle-seconds, and the average integral light value during the first 0.040 seconds of flash of all the charge case loaded assemblies fired for acceptance of each lot shall not be less than 9 million candle-seconds.

3.4 Functioning. The cartridge assembly shall comply with the following requirements.

3.4.1 Dud. The cartridge shall not fail to function and eject the charge case assembly from the launcher.

3.4.2 Premature burst. The charge case loaded assembly shall not explode in the launcher or within a minimum (min.) distance of 50 feet from the launcher.

3.4.3 Delay time. The delay time for each class, from initiation of the cartridge primer to charge case functioning, shall be as specified in Table I.

TABLE I - Delay Time

Class	Minimum (Min.) Seconds	Maximum (Max.) Seconds
1	1.80	2.20
2	3.60	4.40
3	5.40	6.60

3.4.4 Flash and report. After ejection, the cartridge shall not fail to explode with a brilliant flash and a loud report.

3.5 Seal integrity. Each Cartridge, Photoflash, M123A1 Assembly (8847519) shall show no evidence of leakage when tested in accordance with 4.4.3.3.

3.6 Embrittlement of primer shunting clip. The clip shall not break when bent to the minimum point of permanent distortion when tested in accordance with 4.5.7.2.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

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3.8 Workmanship. All parts shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of chips, burrs, sharp edges, cracks, dirt, grease, rust and other foreign material. The metal parts shall not contain metal defects such as inclusions, cold shuts, or porous areas. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the items sample quantities as indicated in Table II.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table II specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 5		DRAWING NUMBER
	Cartridge, Photoflash, M123A1, 2 Sec., 4 and 6 Sec. Delay Assembly				See Below NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Cartridge Case (Dwg. 8847513) Examination for Defects	25		3.2	4.4.2.1
	Charge Case (Dwg. 8847507) Examination for Defects	25		3.2	4.4.2.2
	Cap (Dwg. 8847505) Examination for Defects	25		3.2	4.4.2.3
	Primer Shield (Dwg. 8847511) Examination for Defects	25		3.2	4.4.2.4
	Delay Fuze Housing (Dwg. 8847508) Examination for Defects	25		3.2	4.4.2.5

NOTES:

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 2 OF 5		DRAWING NUMBER
	Cartridge, Photoflash, M123A1, 2 Sec., 4 and 6 Sec. Delay Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze Assembly (Dwg. 8847516) Examination for Defects	25		3.2	4.4.2.6
	Delay Holder (Dwg. 8847510) Examination for Defects	25		3.2	4.4.2.7
	Detonator Holder (Dwg. 8847509) Examination for Defects	25		3.2	4.4.2.8
	Detonator Holder Assembly (Dwg. 8847514) Examination for Defects	25		3.2	4.4.2.9
	Delay Holder Assembly (Dwg. 8847515) Examination for Defects	25		3.2	4.4.2.10

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS MIL-C-12910F(AR)

PARAGRAPH	TITLE: Cartridge, Photoflash, M123A1, 2 Sec., 4 and 6 Sec. Delay Assembly		SHEET 3 OF 5		DRAWING NUMBER: See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Fuze Assembly (2 Second) (Dwg. 8847517)				
	Examination for Defects	25		3.2	4.4.2.11
	Relay Cup (Dwg. 8847501)				
	Examination for Defects	25		3.2	4.4.2.12
	Relay Assembly (Dwg. 8847512)				
	Examination for Defects	25		3.2	4.4.2.13
	Shield Disc Assembly (Dwg. 8847502)				
	Examination for Defects	25		3.2	4.4.2.14
	Primer Shunting Clip (Dwg. 8847506)				
	Examination for Defects	25		3.2	4.4.2.15
	Hardness Test	10		3.2	4.5.7.1
	Embrittlement Test	15		3.2	4.5.7.2

NOTES:

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 4 OF 5		DRAWING NUMBER
	Cartridge, Photoflash, M123A1, 2 Sec., 4 and 6 Sec. Delay Assembly				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Charge Case Loading Assembly (Dwg. 8847518)				
	Examination for Defects	25		3.2	4.4.2.16
	Static Illumination Test	25 a		3.3.2	4.5.1
	Cartridge Assembly (Prior to Inserting Propelling Charge) (Dwg. 8847519)				
	Examination for Defects	25		3.2	4.4.2.17
	Electrical Continuity and Resistance	25 a		4.4.3.5	4.5.5
	Cartridge Assembly (Prior to Inserting Charge Case Loading Assy) (Dwg. 8847519)				
	Examination for Defects	25		3.2	4.4.2.18
	Cartridge Assembly (Dwg. 8847519)				
	Examination for Defects	35		3.2	4.4.2.19
	Functioning	25 a		3.4	4.5.2
	Seal integrity	10 a		3.5	4.5.3

NOTES:
a. To minimize sample quantities these inspections and tests may be performed on the same test items after subsequent assembly operation.

DRSMC-OA (D) Form 160, 1 Aug 83 replaces edition of 1 Jul 77 which may be used until exhausted.

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TABLE II. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE: Cartridge, Photoflash, M123A1, 2 Sec., 4 and 6 Sec. Delay Assembly	SHEET 5 OF 5	DRAWING NUMBER: See Below
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Carton (Prior to Sealing) (Dwg. 7548472) Examination for Defects	10 a		3.2	4.4.2.20
	Sealed Carton (Dwg. 7548472) Examination for Defects	10 a		3.2	4.4.2.21
	Leak Test	10 a	0 - 1	3.2	4.5.8
	Packing Box (Prior to Closing Box) (Dwg. 7548473) Examination for Defects	10 a		3.2	4.4.2.22
	Sealed Packing Box (Dwg. 7548473) Examination for Defects	10 a		3.2	4.4.2.23

NOTES:

a. To minimize sample quantities these inspections and tests may be performed on the same test items after subsequent assembly operation.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of Photoflash Cartridge Assemblies shall contain:

a. Parts of one interfix lot number from one manufacturer.

b. Primers on one interfix lot number from one manufacturer.

c. Composition ingredients from not more than one lot from one supplier.

d. Cartridge assemblies of not more than one class.

4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans procedures for major and minor defects shall be in accordance with MIL-STD-105.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Cartridge Case	1 OF 1	8847513
			NEXT HIGHER ASSEMBLY: 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Inside diameter, min.		100%	3.2	Gage
Major					
101	Concentricity of outside diameter with thread		0.40%	3.2	Gage
102	Diameter of flash hole		0.40%	3.2	Gage
103	Crack or split		0.40%	3.2	Visual
Minor					
201	Total length		0.65%	3.2	Gage
202	Wall thickness, min.		0.65%	3.2	Gage
203	Inside diameter, max.		0.65%	3.2	Gage
204	Pitch diameter of thread,		0.65%	3.2	Gage
205	Minor diameter of thread, max.		0.65%	3.2	Gage
206	Thickness between base and bottom of expelling charge cavity		0.65%	3.2	Gage
207	Diameter of expelling charge cavity, min.		0.65%	3.2	Gage
208	Depth of primer cavity		0.65%	3.2	Gage
209	Thickness from base to forward end of expelling charge cavity		0.65%	3.2	Gage
210	Porous area or inclusion		0.65%	3.2	Visual
211	Chamfer missing		0.65%	3.2	Visual
212	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Charge Case				8847507
					NEXT HIGHER ASSEMBLY
					8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Crack or split		100%	3.2	Visual
Major	None defined				
Minor					
201	Length		0.65%	3.2	Gage
202	Inside diameter		0.65%	3.2	Gage
203	Pitch diameter of thread		0.65%	3.2	Gage
204	Minor diameter of thread		0.65%	3.2	Gage
205	Outside diameter		0.65%	3.2	Gage
206	Thickness through bottom		0.65%	3.2	Gage
207	Length to disc support boss		0.65%	3.2	Gage
208	Wall thickness, min.		0.65%	3.2	Gage
209	Outside diameter, max.		0.65%	3.2	Gage
210	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Cap				8847505
					NEXT HIGHER ASSEMBLY
					8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major					
101	Crack or split		0.40%	3.2	Visual
Minor					
201	Diameter at case contact surface		0.65%	3.2	Gage
202	Outside diameter, min.		0.65%	3.2	Gage
203	Length of sides		0.65%	3.2	Gage
204	Bare spot in protective coating or coating missing		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.4	Primer Shield	1	1	8847511
				NEXT HIGHER ASSEMBLY: 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	One or more holes missing		0.40%	3.2	Visual
Minor					
201	Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Protective finish		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Delay Fuze Housing	1 OF 1	8847508
			NEXT HIGHER ASSEMBLY 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METH
Critical	None defined				
Major	None defined				
Minor					
201	Pitch diameter of thread		0.65%	3.2	Gage
202	Major diameter of thread, min.		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Diameter of hole		0.65%	3.2	Gage
205	Diameter below flange		0.65%	3.2	Gage
206	Diameter of flange		0.65%	3.2	Gage
207	Length from flange over thread		0.65%	3.2	Gage
208	Length of thread undercut, max.		0.65%	3.2	Gage
209	Thickness of flange		0.65%	3.2	Gage
210	Depth of hole counterbore		0.65%	3.2	Gage
211	Length of crimping lip		0.65%	3.2	Gage
212	Diameter of hole counterbore		0.65%	3.2	Gage
213	Protective coating missing or contains bare spot		0.65%	3.2	Visual
214	Wrench flat missing		0.65%	3.2	Visual
215	Chamfer missing		0.65%	3.2	Visual
216	Evidence of poor workmanship		1.0%	3.8	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Fuze Assembly (4 and 6 seconds)				8847516
					NEXT HIGHER ASSEMBLY 8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Charge more than max. below flush		100%	3.2	Gage
2.	Explosive component missing		100%	3.2	Visual
Major					
101	Marking or stamping misleading or unidentifiable		0.40%	3.2	Visual
102	Crimp missing or not full 360 degrees		0.40%	3.2	Visual
Minor					
201	Charge above flush		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.7	Delay Holder		8847510
			NEXT HIGHER ASSEMBLY 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flash hole missing or obstructed		0.40%	3.2	Visual
Minor					
201	Pitch diameter of charge case thread		0.65%	3.2	Gage
202	Major diameter of charge case thread		0.65%	3.2	Gage
203	Pitch diameter of detonator holder thread		0.65%	3.2	Gage
204	Major diameter of detonator holder thread		0.65%	3.2	Gage
205	Length		0.65%	3.2	Gage
206	Perpendicularity of detonator holder seating surface with detonator holder thread		0.65%	3.2	Gage
207	Length of detonator holder thread, min.		0.65%	3.2	Gage
208	Length of thread to flange		0.65%	3.2	Gage

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MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.7	Delay Holder	2 OF 2	8847510
			NEXT HIGHER ASSEMBLY 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
209	Length of thread undercut, max.		0.65%	3.2	Gage
210	Diameter of hole		0.65%	3.2	Gage
211	Large inside diameter		0.65%	3.2	Gage
212	Thickness of flange		0.65%	3.2	Gage
213	Thickness through bottom of cavity		0.65%	3.2	Gage
214	Diameter of flange		0.65%	3.2	Gage
215	Large outside diameter between shoulder and flange		0.65%	3.2	Gage
216	Length to shoulder		0.65%	3.2	Gage
217	Protective coating missing or contains bare spot		0.65%	3.2	Visual
218	Wrench flat missing		0.65%	3.2	Visual
219	Chamfer missing		0.65%	3 2	Visual
220	Evidence of poor workmanship		1.0%	3.3	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Detonator Holder		8847509
			NEXT HIGHER ASSEMBLY 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Flash hole missing or obstructed		0.40%	3.2	Visual
Minor					
201	Length		0.65%	3.2	Gage
202	Length of crimping lip		0.65%	3.2	Gage
203	Depth of detonator cavity		0.65%	3.2	Gage
204	Depth of thread cavity		0.65%	3.2	Gage
205	Length of thread, min.		0.65%	3.2	Gage
206	Pitch diameter of thread		0.65%	3.2	Gage
207	Minor diameter of thread, max.		0.65%	3.2	Gage
208	Depth of thread counterbore		0.65%	3.2	Gage
209	Diameter		0.65%	3.2	Gage
210	Wrench flat missing		0.65%	3.2	Visual
211	Chamfer missing		0.65%	3.2	Visual
212	Protective coating missing or contains bare spot		0.65%	3.2	Visual
213	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.9	Detonator Holder Assembly	1 OF 1	8847514
			NEXT HIGHER ASSEMBLY: 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Detonator inverted or missing		0.40%	3.2	Visual
102	Crimp missing or not full 360 degrees		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Delay Holder Assembly				8847515
					NEXT HIGHER ASSEMBLY
					8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Charge more than max. below flush		100%	3.2	Gage
Major					
101	Marking or stamping misleading or unidentifiable		0.40%	3.2	Visual
Minor					
201	Charge above flush		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Fuze Assembly (2 Second)				8847517
					NEXT HIGHER ASSEMBLY 8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Cement missing		0.40%	3.2	Visual
Minor 201	Delay holder assembly loose on detonator holder assembly		0.65%	3.2	Manual
202	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.12	Relay Cup		8847501
			NEXT HIGHER ASSEMBLY 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Crack or split		100%	3.2	Visual
Major	None defined				
Minor					
201	Diameter		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Length		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.13	Relay Assembly	1 OF 1	8847512
			NEXT HIGHER ASSEMBLY 8847519

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Charge above flush or more than max. below flush		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Shield, disc assembly				8847502
					NEXT HIGHER ASSEMBLY
					8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Paper disc missing, perforated, or torn		0.65%	3.2	Visual
202	Adhesive over hole		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.15	Primer shunting clip				8847506
					NEXT HIGHER ASSEMBLY
					8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile of ends		0.40%	3.2	Gage
Minor					
201	Length between contact surfaces		0.65%	3.2	Gage
202	Thickness		0.65%	3.2	Gage
203	Width		0.65%	3.2	Gage
204	Width over center contacts		0.65%	3.2	Gage
205	Thickness of center contact		0.65%	3.2	Gage
206	Protrusion of center locking tabs from contact		0.65%	3.2	Gage
207	Protective coating missing or contains bare spot		0.65%	3.2	Visual
208	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.16	Charge Case Loading Assembly				8847518
					NEXT HIGHER ASSEMBLY
					8847519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION ME
Critical					
1.	Crimp not flat against closing disc		100%	3.2	Visual
2.	Fuze assembly loose		100%	3.2	Manual
3.	Diameter after crimping, max.		100%	3.2	Gage
Major					
101	Sealing cement missing or inadequate		0.40%	3.2	Visual
102	Delay marking missing or incorrect		0.40%	3.2	Visual
103	Static illumination		4.4.3.1	3.3	4.5.1
Minor					
201	Marking, other than delay marking, missing or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.17	Cartridge Assembly (Prior to Inserting Propelling Charge)				8847519
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	Primer above flush		100%	3.2	Visual/Straight-Edge
Major					
101	Depth to primer, max.		0.40%	3.2	Gage
102	Sealing composition missing or where not required		0.40%	3.2	Visual
103	Electrical continuity and resistance		100%	3.2	4.4.3.5
Minor					
201	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.18	Cartridge Assembly (Prior to Inserting Charge Case Loading Assembly)		8847519
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Loose explosive in case		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.0%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.19	Cartridge Assembly				8847519
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	X-ray performance test symbol missing		100%	3.2	Visual
2.	Primer shunting clip missing, loose or does not contact primer electrode		100%	3.2	Visual
3.	Cartridge case dented		100%	3.2	Visual
4.	Seal integrity		100%	3.5	4.5.3
Major					
101	Length over clip, max.		0.40%	3.2	Gage
102	Contents loose		0.40%	3.2	Manual
103	Crimp of cover not full 360 degrees		0.40%	3.2	Visual
104	Sealing compound missing, inadequate or present where not required		0.40%	3.2	Visual
105	Delay marking missing or incorrect		0.40%	3.2	Visual
106	Functioning		4.4.3.2	3.4	4.5.2
Minor					
201	Marking, other than delay marking, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.0%	3.8	Visual

NOTES:

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QUALTIY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548472
4.4.2.20	Carton (Prior to Sealing)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box cut, punctured or torn in any area		0.40%	3.2	Visual
Minor					
201	Any packing material missing		0.65%	3.2	Visual
202	Incorrect number of cartridges		0.65%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.21	Sealed Carton				7548472
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag punctured, torn or seal incomplete		0.40%	3.2	Visual
102	Quick leak test	10/ Cart	1-2	4.4.3.8	4.5.8
Minor					
201	Contents loose		0.65%	3.2	Manual
202	Marking misleading or unidenitifable		0.65%	3.2	Visual
203	Label missing or only partially sealed (when applicable)		0.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548473
4.4.2.22	Packing Box (Prior to Closing)				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Any packing material missing		0.40%	3.2	Visual
202	Incorrect number of boxes		0.40%	3.2	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-12910F(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.23	Sealed Packing Box				7548473
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1.	X-ray performance test stamp with date missing		100%	3.2	Visual
Major					
101	Board broken or split so that contents are exposed or liable to become exposed		0.40%	3.2	Visual
102	Hardware or strapping broken or loose		0.40%	3.2	Visual-Manual
103	DOT nomenclature missing or incorrect		0.40%	3.2	Visual
Minor					
201	Hardware or strapping improperly engaged		0.65%	3.2	Visual
202	Metal seal missing, unsealed or mislocated		0.65%	3.2	Visual-Manual
203	Contents loose		0.65%	3.2	Manual
204	Marking misleading or unidentifiable		0.65%	3.2	Visual

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4.4.3 Testing.

PRECAUTION: This specification covers sampling and testing of toxic and hazardous materials. Accordingly, it is emphasized that all applicable safety rules regulations and procedures must be followed in handling and processing the item.

4.4.3.1 Static Illumination of Charge Case Loaded Assembly (See 3.3) Major Defect. The charge case loading assemblies shall be observed for duds, integral light value of each charge loaded assembly and the average integral light value of all loaded charge case assemblies. Charge case loaded assemblies may be selected from cartridge assemblies that have been subjected to the seal integrity test except that charge cases shall not be used from cartridge assemblies that leaked.

4.4.3.1.1 First lot produced. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, a sample of 50 charge case assemblies shall be selected from each lot for this test. If the average of the integral light values of the 50 charge case loaded assemblies tested is less than the requirement specified, or if 3 or more charge case loaded assemblies fail to fire or exhibit a reading of less than the minimum value specified, the lot shall be rejected. The test shall be performed as specified in 4.5.1.

4.4.3.1.2 After three (3) consecutive lots. After three (3) consecutive lots have met the criteria of 4.4.3.1.1, a sample of 20 charge case loaded assemblies shall be selected from each lot for test. If the average of the integral light value of the 20 charge case loaded assemblies is less than the requirement specified, or if 2 or more assemblies tested, fail to fire or exhibit a reading of less than the minimum value specified, the lot shall be rejected.

4.4.3.2 Functioning (see 3.4). The cartridge assembly shall be observed for evidence of the following defects:

a. Premature burst - Critical defect
b. Dud - Major defect
c. Delay time not within tolerance - Major defect
d. Flash and report obviously inadequate- Major defect

4.4.3.2.1 First lot produced. Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, one hundred and twenty five (125) samples shall be selected for this test. The lot shall be rejected if a Critical defect occurs at any time, or if three (3) or more samples show evidence of a Major defect. The test shall be performed as specified in 4.5.2.

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4.4.3.2.2 After three (3) consecutive lots. After three (3) consecutive lots have met the criteria of 4.4.3.2.1, fifty (50) samples shall be selected from each lot for test. The lot shall be rejected if a Critical defect occurs at any time, or if two (2) or more samples show evidence of a Major defect.

4.4.3.3 Seal integrity (see 3.5) - Critical defect - (Non-destructive test) This test shall be specified in 4.5.3. Any assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.4.3.4 Moisture content (see Table III) - (See dwgs. 8847512, 8847515, 8847516, 8847518, 8847519) - Major defects

TABLE III - Moisture Content

Material
Delay composition
Lead azide
Igniter composition
Lead styphnate
Black powder
Photoflash powder ingredients
Felt
Paper
Chipboard

The contractor shall provide adequate controls to insure compliance with the requirements, and shall test for verification, at least one sample of each charge, composition, and material from each eight (8) hours of production of cartridges. Composite samples shall not be used. If the moisture content of a sample exceeds the requirement, and loading has not begun, that quantity of material represented by the sample, shall be rejected. If assemblies have been loaded with material containing excessive moisture, the remaining material that is represented by the sample together with all assemblies loaded with non-conforming material shall be rejected. The tests shall be performed as specified in 4.5.4.

4.4.3.5 Electrical continuity and resistance (see dwg. 8847519) - Major defect. This test shall be performed 100 percent in accordance with the procedure specified in 4.5.5. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.4.3.6 Determination of composition ingredients (see Table IV) - (See dwgs. 8847515, 8847516 and 8847518) - Major defects.

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TABLE IV - Determination of Composition

Material
Delay composition
Igniter composition
Photoflash composition

A sample of not less than 5 grams of each composition shall be selected from each batch or mixture for the test. If any composition fails to comply with the dwg. requirements, the lot/batch shall be rejected. When blending is accomplished in individual charge cases rather than by batch blending, the adequacy of the composition may be checked by emptying one or more of the partially mixed charges from the charge cases before they have been closed, blending the ingredients and performing an analysis on the sample of the mixture. The test shall be performed as specified in 4.5.6.

4.4.3.7 Hardness and embrittlement of primer shunting clip (see dwg. 8847506) - Major defect. Twenty (20) primer shunting clips shall be selected from each lot for tests. If any clip fails to comply with the requirements, the lot shall be rejected. The test shall be performed as specified in 4.5.7.

4.4.3.8 Quick leak test (see dwg. 7548472) - Major defect. Ten (10) cartons in sealed barrier bags shall be selected from each lot for test. If two (2) or more cartons show evidence of leakage, the lot shall be rejected. The test shall be performed as specified in 4.5.8.

4.4.3.9 Radiographic inspection of cartridge assemblies (see dwg. 8847519). Each cartridge assembly in the lot shall be subjected to 100 percent radiographic examination in accordance with the provisions of 4.5.9. Each radiograph shall be examined for evidence of the following defects, using the radiographic standards prepared by the contractor and submitted for approval.

a. Less than 80 percent of propelling charge present (Critical defect)
b. Delay column missing (Critical defect)
c. Shield disc assembly missing (Critical defect)
d. Gas check wad missing (Critical defect)
e. Fuze gasket missing (Critical defect)
f. Crimp of closing disc less than 90 degress (Critical defect)
g. Relay assembly missing (Major defect)

All assemblies containing one or more of the defects listed herein shall be classed defective and removed from the lot. During radiograph examination, if more than two (2) percent of the lot is found to contain more than 80 percent but less than 90 percent of the propelling charge, the lot shall be rejected subject to segregation.

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4.4.3.10 Check test for deterioration of Primer, Electric, M59 (see MIL-P-11329). If the total elapsed time between original acceptance of the M59 Primer lot and the assembly of that lot into the cartridge assembly exceeds two years; or if the M59 primers have been subjected to adverse conditions, however brief, at any time since previous tests, the applicable lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in MIL-P-11329. The test shall be performed by the contractor on assemblies selected by the Government inspector, at the facility assembling the item.

4.4.4 Inspection Equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the 'Paragraph Reference/Inspection Method' column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of Inspection.

4.5.1 Static illumination of charge case assembly. The light output of each assembly shall be measured at two points. One point shall be the end of the fuze and the other shall be 90 degrees from this point. The procedure and equipment shall be as specified on dwg. 9278637. Observation shall be made for compliance with the applicable requirement. (Destructive Test)

4.5.2 Functioning. The cartridges, with primer end up, shall be securely placed in a holding fixture so as to prevent any movement within the fixture. The holding fixture shall be placed in a conditioning box and vibrated horizontally at 100 cycles per second and an amplitude of 0.020 in. at a temperature of minus 30 $\pm$ 5°F, (minus 34 $\pm$ 2°C) for 2 hours. Within 5 minutes of removal from conditioning chamber, the cartridges shall be fired from a ground launcher. The delay time shall be determined by a timing system accurate within plus or minus 0.01 second. The cartridges shall be observed for compliance with the requirements. (Destructive Test)

4.5.3 Seal integrity. Each Cartridge Assembly (8847519) shall be placed in a cylindrical air tight chamber with the free space minimized. The air pressure (moisture free) within the chamber shall be raised to a minimum of 3 p.s.i.g. by means of a fixed volume of air. With the air supply shut off the pressure shall be maintained for 15 seconds minimum. Any Cartridge Case tested which fails to attain the minimum pressure specified or exhibits any leakage in excess of 0.05 p.s.i. shall be rejected.

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4.5.4 Moisture content determinations.

4.5.4.1 Delay composition. Determine the moisture content of the delay composition in accordance with method 102.1 of MIL-STD-1234 using five (5) g. sample of delay composition.

Percent moisture equals $\frac{100A}{B}$

where:
A = Loss of weight (gm)
B = Weight of original sample (gm)

4.5.4.2 Lead azide. The moisture content of the lead azide shall be determined as specified in 4.5.4.1 except that a 2 gram sample shall be used, and the drying temperature shall be 70 ± 2°C (158 ± 5°F).

4.5.4.3 Igniter composition. The igniter composition moisture content shall be determined as specified in 4.5.4.1.

4.5.4.4 Lead Styphnate. The moisture content of the lead styphnate shall be determined as specified in 4.5.4.1 except that a 2 gram sample shall be used, and the drying temperature shall be 70 ± 2°C (158 ± 5°F).

4.5.4.5 Black powder. The moisture content of the black powder shall be determined in accordance with Specification MIL-P-223, using a sample weight of appproximately 0.5 grams and heating thirty (30) minutes at 70 ± 2°C (158 ± 5°F).

4.5.4.6 Photoflash composition. The moisture content of the photoflash composition shall be determined as specified in 4.5.4.1 except that a 10 g. sample and a drying temperature of 100 ± 2°C (212 ± 5°F) shall be used.

4.5.4.7 Felt. The moisture content of the felt shall be determined as specified in 4.5.4.1 except that the sample size shall be 10 g. and the drying time until a constant weight is attained.

4.5.4.8 Chipboard. The moisture content of the chipboard shall be determined as specified in 4.5.4.1, except that the sample size and drying temperature shall be as specified in 4.5.4.6.

4.5.5 Electrical continuity and resistance. The test for electrical continuity and resistance shall be conducted prior to inserting the propelling charge. Observation shall be made for compliance with the applicable drawing requirement. (Non-Destructive Test)

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4.5.6 Determination of compositions.

4.5.6.1 Delay composition.

4.5.6.1.1 Procedure (Barium chromate).

4.5.6.1.1.1 Perchloric acid solution. Mix 500 milliliters (mL.) of perchloric acid (70 to 72 percent vacuum distilled) with 500 mL. of water.

4.5.6.1.1.2 Ortho-phenanthroline-ferrous sulphate indicator. Dissolve 1.487 g. of ortho-phenanthroline monohydrate in 100 mL. of an aqueous ferrous sulfate solution containing 0.695 g. of ferrous sulfate pentahydrate per 100 mL.

4.5.6.1.1.3 Standard ferrous perchlorate. Add 129 mL. of perchloric acid (70 to 72 percent vacuum distilled) to 871 mL. or water; add 12 g. of a pure grade of iron powder and stir the solution until the iron is completely dissolved. Filter the resultant solution through a 200 ml. medium porosity sintered glass crucible. Pass a stream of nitrogen through the solution for 2 hours before using.

4.5.6.1.1.4 Standardization of ferrous perchlorate. Standardize the ferrous perchlorate solution against potassium dichromate of known purity, such as Bureau of Standards, 136 in the following manner:

Transfer a portion of approximately 0.5 g. of the potassium dichromate to a 500 mL. Erlenmeyer flask, add 50 mL. of water and stir the mixture until the salt is completely dissolved. Add 50 mL. of perchloric acid solution and titrate the solution with the ferrous perchlorate solution, by adding the ferrous perchlorate solution until the solution being titrated acquires a greenish-blue color. Add 2 drops of ortho-phenanthroline-ferrous sulfate indicator and stir the solution until the pink color disappears.

Continue the titration until the color of the solution changes from a greenish-blue to a pink color which lasts for approximately 2 minutes. Calculate the normality of the ferrous perchlorate solution as follows:

$$\text{Normality of the ferrous perchlorate solution} = \frac{AW}{4.904V}$$

where:

A = purity percentage of the potassium dichromate
W = weight of potassium dichromate used, g.
V = ferrous perchlorate used, mL.

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4.5.6.1.1.5 Percentage of barium chromate. Transfer an accurately weighed portion of approximately 1 g. of the sample to a 300 mL. Erlenmeyer flask. Place the flask and contents on a cold hot plate, and carefully add 25 mL. of perchlorate acid by means of a pipette so that the acid runs down the side of the flask and does not spatter the sample, since boron adhering to the sides as a result of spattering does not give precise results. Heat the flask with medium heat until the boron is completely dissolved and fumes of perchloric acid are given off. Continue heating for 10 additional minutes after red chromic anhydride precipitates out of the solution. Remove the flask from the hot plate, add 75 mL. of distilled water and cool rapidly to room temperature in an ice bath. Pass a stream of nitrogen through the solution for 5 minutes before titrating the solution with the standard ferrous perchlorate solution. Add the ferrous perchlorate solution rapidly stirring vigorously and titrate the solution until the latter solution assumes a greenish-blue color. Add 2 drops of ortho-phenanthroline-ferrous sulfate indicator and stir until the pink color disappears. Continue the titration until the color of the solution changes from a greenish-blue to a pink or red color. Calculate the percentage of barium chromate in a sample as follows:

$$\text{Percentage of barium chromate} = \frac{8.446W}{WP}$$

where:

V = mL. of standard ferrous perchlorate solution used.
W = weight of the sample, g.
N = normality of the standard ferrous perchlorate solution.
P = Purity of barium chromate used. The purity of barium chromate shall be determined for each lot of barium chromate used in delay compositions.

4.5.6.1.1.6 Percentage of boron. Calculate the percentage of boron in the sample by subtracting the percentage of barium chromate from 100.

4.5.6.2 Igniter composition. The procedure for determining the igniter compositon shall be as specified in 4.5.6.1.

4.5.6.3 Photoflash composition

4.5.6.3.1 Aluminum. Transfer an accurately weighed portion of approximately two (2) g. of the sample to a tared medium porosity sintered glass crucible. Extract the sample with ten 25 mL. portions of distilled water at a temperature of 0 to + 5°C, (32 + 10°F), using a filtering flask and suction. Stir the contents of the crucible with each portion of water for 1 minute before applying suction. Transfer the extract quantitatively to a 400 mL. beaker

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using approximately 50 mL. of wash water and reserve the extract and washings for the barium nitrate determinations. Rinse the undissolved residue with alcohol and ether, and aspirate until no further solvent odors are apparent. Place the crucible and contents in a drying oven at 100 + 5°C (212 + 10°F) for 1 hour, cool in a desiccator and weigh. Calculate the increase in weight of the crucible to percent of aluminum powder on a moisture free basis as follows:

$$\text{Percentage of aluminum powder} = \frac{100\ A}{W-WM}$$

where:
A = Increase in weight of gooch crucible
W = Weight of sample.
M = percent moisture expressed as a decimal

4.5.6.3.2 Barium nitrate. To the extract reserved during the aluminum powder determination, add 1 mL. of 12N hydrochloric acid. Heat to boiling on a hot plate and add slowly from a 50 mL. pipette, 50 mL. of 3N (1:8) sulfuric acid. Allow the mixture to boil gently for five minutes and finally to digest for at least 2 hours on a steam bath. Collect the precipitated barium sulfate in a tared Gooch crucible which has been previously heated in a muffle furnace at approximately 750°C (1382°F), for 30 minutes. Wash the precipitate 10 times with 25 mL. portions of hot water, dry in an oven at 100 ± 5°C, (212 ± 10°F) for 1 hour and in a muffle furnace at approximately 750°C, (1328°F) for 30 minutes, cool in a desiccator and weigh. Calculate the percentage of barium nitrate in the sample on a moisture free basis as follows:

$$\text{Percentage of barium nitrate} = \frac{111.98A}{W-WM}$$

where:
A = increase in weight of Gooch crucible, g.
W = weight of sample, g.
M = percent moisture expressed as a decimal

4.5.6.3.3 Potassium perchlorate. Calculate the percentage of potassium perchlorate in the sample by deducting from 100 the percentage of aluminum and barium nitrate found.

4.5.7 Primer shunting clip.

4.5.7.1 Hardness. The shunting clip shall be tested as specified in Method 243 of Federal Standard 151. Observation shall be made for compliance with the applicable drawing requirement.

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4.5.7.2 Embrittlement of primer shunting clip. The clip shall be bent to the minimum point of permanent distortion by any method satisfactory to the procuring activity. All shunting clips subjected to the test shall be scrapped. Observation shall be made for compliance with the applicable requirement. (Destructive Test)

4.5.8 Quick leak test. The test shall be performed in accordance with the procedure in Specification MIL-P-116, using the vacuum chamber or the pneumatic pressure technique. (Non-Destructive Test)

4.5.9 Radiographic inspection of cartridge assembly. The cartridge assemblies shall be subjected to radiography and examination in accordance with the radiographic standards. (Non-Destructive Test)

5. PACKAGING

5.1 Preservation an packaging.

5.1.1 Level A. The cartridge shall be preserved and packaged in accordance with dwg. 7548472.

5.2 Packing.

5.2.1 Level A. The boxes shall be packed in accordance with dwg. 7548473.

5.3 Marking. Marking shall be as specified on dwgs. 7548472 and 7548473.

5.4 Unitization. Unitization shall be in accordance with Dwg. 19-48-4116/106A.

5.5 Lot integrity. Lots shall not be intermingled for shipment. Division between lots in a shipment shall be distinctly identified.

6. NOTES

6.1 Intended use. The components covered by this specification are intended for use on the Cartridge, Photoflash, M123A1.

6.2 Ordering data. See MIl-A-48078

6.3 Submission of inspection equipment for design approvals. See MIL-S-48078. Submit designs as required to Commander, US Army Armament Research and Development Center, ATTN: DRSMC-QAT-I (D), Dover, NJ 07801.

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6.4 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

6.5 Equivalent test methods. The test methods given in this specification are the official methods to be used. The contractor may request using other methods providing that the proposed method is equivalent (accuracy and precision) to the method given in this specification. Prior approval of the Contracting Officer is required for use of equivalent test methods. A description of the proposed method should be submitted through the Contracting Officer to: Commander, ARDC, ATTN: DRSMC-QAT-M (D), Dover, NJ 07801. This description should include, but not be limited to, the procedures used, the accuracy and precision and drawings of any special equipment required (see MIL-I-45208, paragraph 3.10).

6.6 Safety data. For Safety Data Requirements refer to Safety Data Documents Hazardous Component Safety Data Statement No. 0169.

6.7 Changes from Previous Issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

6.8 Identification of international standardization agreement. Certain provisions of this specification (identified by paragraph number of similar manner, if appropriate) are the subject of international standardization agreement (insert the ABCA or ASCC standard number, the NATO, NETR or NEPR number, or other appropriate documentary reference). When amendment, revision, or cancellation of this specification is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental standardization offices to change the agreement or make other appropriate accommodations.

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(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-12910F	CARTRIDGE, PHOTOFLASH, M123A1, PARTS FOR, AND LAP

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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